PORTFOLIO CASE STUDY · FINANCIAL SERVICES

Customer Segmentation & Retention Analytics

Segmenting 4,000 fictional retail-banking customers by relationship value and engagement, and finding the early signs that they are about to leave



Ahsan Anas · October 2026

All customers, data and results are synthetic. Built with AI assistance (Claude) under the author's direction – see README.

THE BUSINESS PROBLEM

Which customers create value, which are drifting away, and what should we do about it?

A retail bank's customer book is rarely one market. High-value customers can be quietly disengaging, while highly engaged customers may hold only one product. Treating them all the same wastes retention budget and misses growth.

Questions this project answers

- Who are our customer segments, and how much value does each hold?
- Which behaviours separate engaged from drifting customers?
- Which warning signals predict attrition – and how much revenue is at risk?
- Which interventions are worth testing, and how would we test them?

1. **Generate**
   Seeded simulation of 4,000 fictional customers with linked behaviours, plus realistic data errors
2. **Clean & validate**
   14 logged checks; nothing imputed; 39/39 automated checks pass
3. **Segment**
   Revenue proxy × Relationship Engagement Index → six actionable segments, stress-tested
4. **Retain & grow**
   Warning signals, an attrition model, sized hypothetical interventions and a pilot design

Labels used throughout: SYNTHETIC OBSERVATION · MODEL ASSUMPTION · HYPOTHETICAL

4,085 raw rows became 4,000 clean customer records – and no value was invented

4,000 fictional customers, 39 fields: demographics, balances, products, transactions, digital, service

14 logged cleaning checks catching every injected issue type (681 injected problems)

0 values imputed – blanks stay blank and scores use only the fields available

39/39 automated validation checks pass (reproducibility, integrity, calculations, Excel = Python)

Issue found	Rows	Action
Exact duplicates + re-ingested older snapshots	85	Removed duplicates; kept each customer's newest record
Region / employment coding variants	125	Standardised ('METRO', 'Metropolitan' → Metro)
Mixed date formats (DD/MM/YYYY vs ISO)	140	Parsed each format explicitly – no guessing
Income recorded in thousands (e.g. 54.3)	36	×1,000 unit fix, flagged
Impossible ages, sentinel tenure (−1), future dates	40	Set to missing and flagged
Missing income · card limit	161	Left missing – not imputed
Negative savings, stale product counts, 900% card use	92	0 if no product else missing; recount from flags; set missing

Every rule is in reports/data_quality_log.csv and is checked against the generator's list of injected issues.

SEGMENTATION MODEL

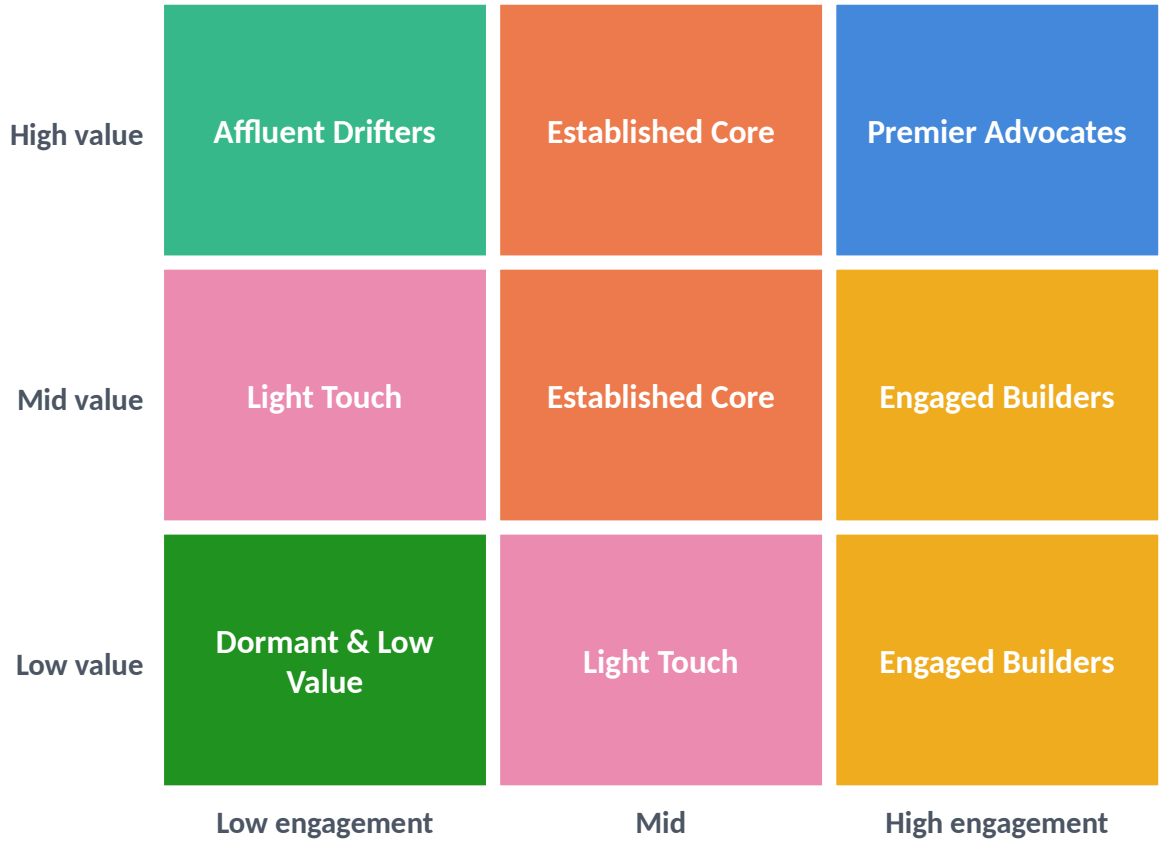
Two axes – what a customer is worth and how engaged they are – give six segments with a clear action each

MODEL ASSUMPTION

Value axis: estimated annual revenue
Balances × illustrative margins (current account 2.5%, savings 1.2%, mortgage 1.1%, loans 4.5%, investments 0.75% fee) + card interchange. Bands: High ≥ $1,500 · Mid ≥ $400.

Engagement axis: Relationship Engagement Index (0–100)
Recency 30 (halves every 30 days) · frequency 20 · digital 15 · product adoption 20 · loyalty 15. Bands: High ≥ 70 · Mid ≥ 50.

Stress-tested
Five alternative weightings keep 84–90% of customers in the same segment; value cut-offs ±20%: 93–94%; engagement cut-offs ±5 points: 79–80% – the engagement cut-off is the most sensitive choice. k-means finds little natural structure (silhouette 0.20–0.22; agreement ARI 0.18), so transparent rules were kept: each segment maps to an action.

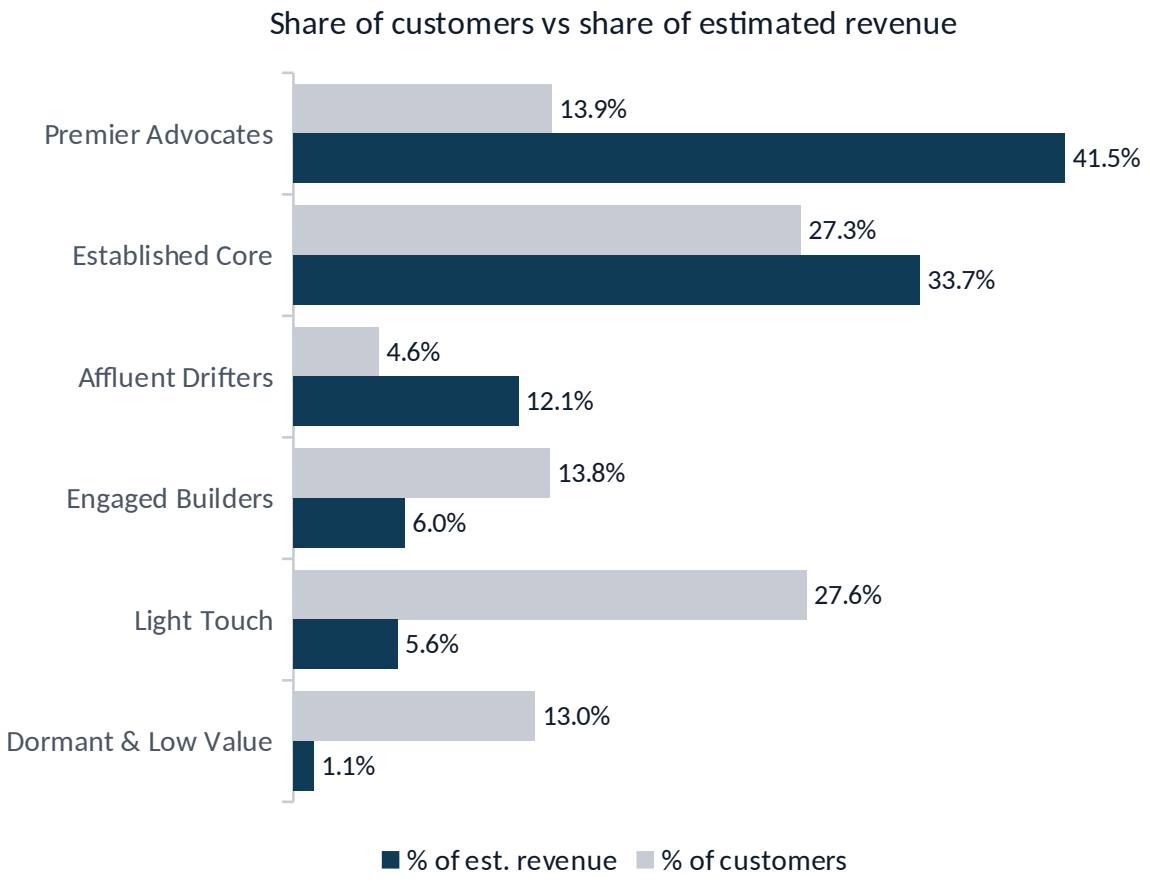


Rule table in src/config.py; rebuilt with live formulas in the Excel workings.

KEY FINDINGS

Value is concentrated, and the costliest risk sits with wealthy customers who have gone quiet

SYNTHETIC OBSERVATION



Premier Advocates
14% of customers hold 42% of revenue; attrition only 2.2%. Protect and reward.

Affluent Drifters
4.6% of customers, 12% of revenue – median 162 days since a transaction and 28% attrition.

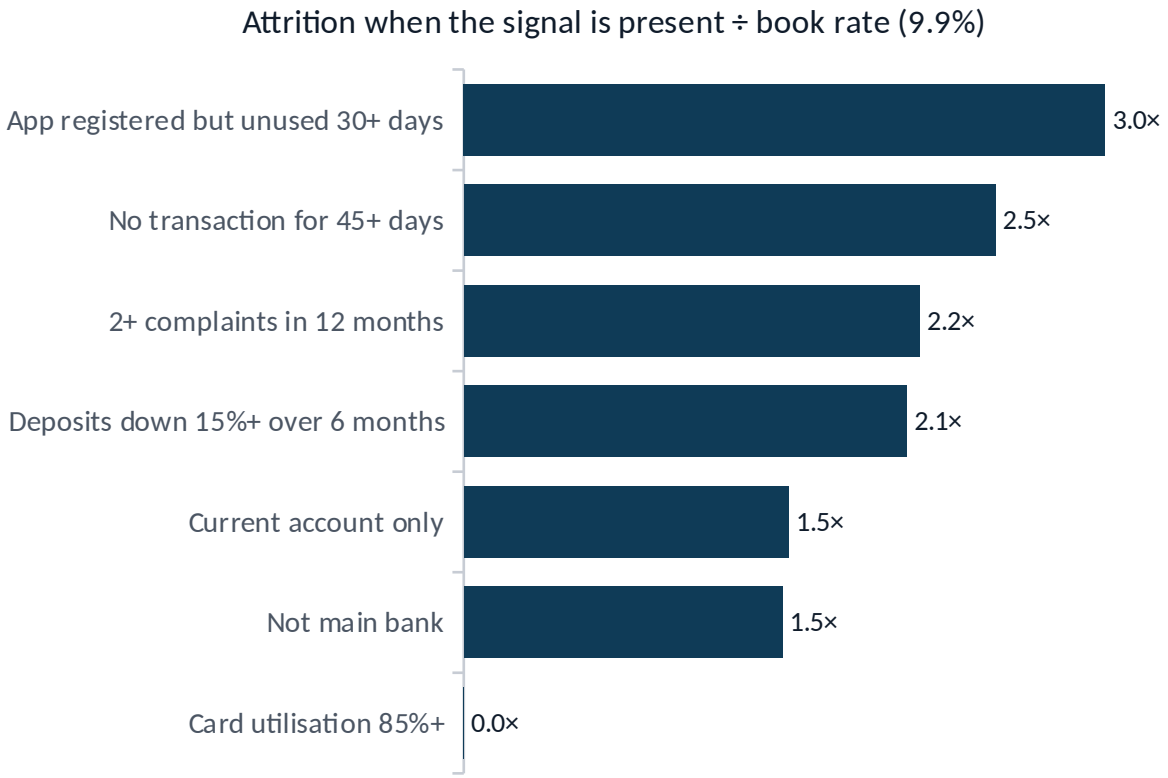
Engaged Builders
Most engaged (REI 76; 84% main bank) but median revenue only $687 – the cross-sell pool.

Light Touch + Dormant
41% of customers but 7% of revenue – low-cost digital servicing.

RETENTION WARNING SIGNALS

400 customers flagged as high risk; $533k of annual revenue is at risk (expected)

SYNTHETIC OBSERVATION



High card utilisation flags only 5 customers and adds no risk here – not every intuitive signal earns its place.

MODEL ASSUMPTION

0.81

test AUC – logistic regression on 9 engagement and value features, 30% held-out test set

3.8×

attrition rate in the top 10% of risk scores vs the average

34%

of revenue at risk sits with Affluent Drifters ($181k), 33% with Established Core

Read with care: the outcome was simulated from the same hidden traits as the features, so this proves the method, not real-world accuracy.

RETENTION AND GROWTH OPPORTUNITIES

Three interventions worth testing – sized on stated assumptions, then proven with a holdout pilot

HYPOTHETICAL

Affluent Drifters

Relationship-manager re-engagement call + needs review

$45k

Expected annual revenue protected (USD)

Assumption: Outreach cuts their attrition probability by 25% (relative)

Engaged Builders

Targeted cross-sell of one savings or card product

$10k

Incremental annual revenue (USD)

Assumption: 10% take up one product worth USD 180 a year

Light Touch

Digital activation (app onboarding, e-statements, bill-pay set-up)

≈5

Customers retained (expected)

Assumption: 15% become regular app users; their model risk falls to that of current regular app users in the segment

Proposed pilot: random 50/50 treatment vs holdout

Cell	Treatment / holdout	Baseline	Smallest detectable change*
Affluent Drifters RM outreach	91 / 91	28.4%	±18.7 pts
Light Touch digital activation	548 / 547	10.6%	±5.2 pts
Engaged Builders cross-sell	273 / 273	5.0%	±5.2 pts

** 95% confidence, 80% power; indicative. The Affluent Drifters cell is small, so only a large effect would show – pool cells or run longer.*

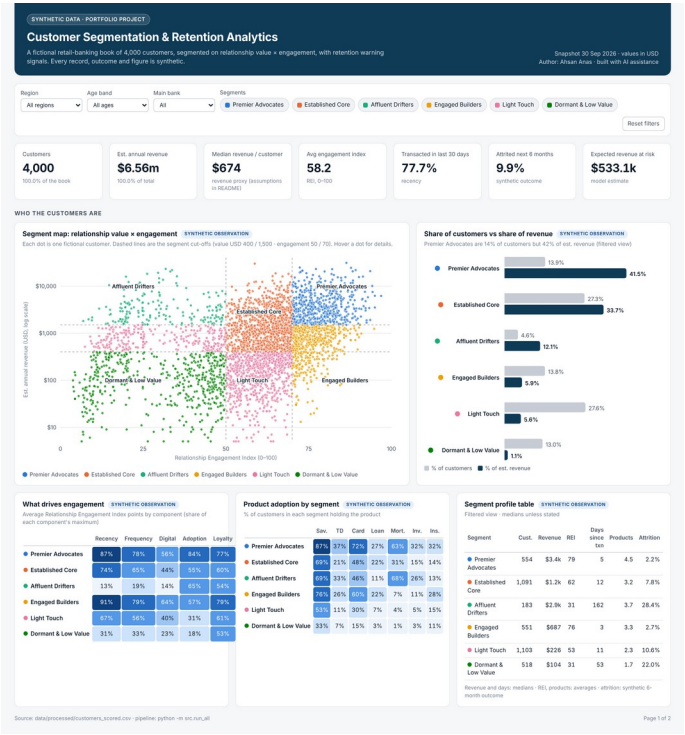
A reusable framework – to be validated on real, governed data before any decision

Limitations

- Synthetic data: relationships exist because the generator created them
- Revenue proxy relies on illustrative margins, not real pricing
- Segment cut-offs are judgement calls (tested, but still choices)
- Attrition model is associative, not causal; untested on real customers
- Scenario values are assumption-driven, not forecasts

Recommendations

- Re-run the pipeline on governed real data; recalibrate margins with Finance
- Prioritise Affluent Drifters for relationship-manager outreach – via a holdout pilot
- Use Engaged Builders as the cross-sell test pool
- Monitor the four strongest warning signals monthly; re-fit the model on observed outcomes
- Add fairness and privacy review before any model drives customer treatment



Interactive dashboard: dashboard/index.html (filters by region, age, main bank and segment)

About this project: designed and directed by Ahsan Anas; code, synthetic data and documents generated with AI assistance (Claude) and reviewed by the author. Reproduce everything with python -m src.run_all.